ASSIGNMENT · AI FOR FIELD SALES

# Build a proof of concept that plans every Sales Officer's month

A laminates manufacturer sells through dealers. Its Sales Officers (SOs) plan a monthly Permanent Journey Plan (PJP) of about 100 focus accounts by hand, and most of it does not happen. Build a working POC that writes a better November 2026 plan for three real (anonymised) SOs, routes each day, nudges the SO before visits, and re-plans when the day goes wrong.

## 1 The problem today

- **Plans are built by hand** from sales history and local knowledge, then keyed into Salesforce (SFDC). Nobody checks whether the plan is the right one.
- **Plans are not followed.** Planned-visit adherence on focus accounts averages about 60%. In the sample data, September adherence was **55% for SO-A and 35% for SO-B**, and SO-C entered no planned visits at all after August.
- **Ad-hoc visits outnumber planned ones** (50–100% of completed visits in the sample). Daily call quotas are met on easy, non-focus accounts: one SO logged 180 calls in a month but covered only 40% of focus accounts.
- **High-value dealers who are de-growing are not seen early in the month.** The plan sees only the company's own sales, so it cannot tell a large dealer giving it a small share from a small dealer giving it everything.
- **Gaming goes unflagged:** repeated visits to the same easy dealer (SO-C has dealer-months with up to 33 visits) and calls under 10 minutes. SFDC reminders are ignored; WhatsApp messages are read.

## 2 What to build: five stages

| 1 · Rank | 2 · Schedule | 3 · Route | 4 · Nudge | 5 · Re-plan |
|---|---|---|---|---|
| Score every dealer. Pick each SO's focus accounts. | Spread dealer visits across November's working days. | Order each day's visits from the SO's start point. | WhatsApp talking points before a visit. | Rewrite the rest of the month after a disruption. |

- **Rank.** Prioritise on three signals agreed with the client's sales leadership: **de-growth** (month-on-month, quarter-on-quarter and versus the same period last year, weighted by value lost, not percentage), **potential** (the dealer's total turnover from GST filings against what it buys from us: headroom) and **sudden drops**. Use loyalty-club status and distance to the next scheme slab where useful. Output a ranked list and each SO's focus dealers.
- **Schedule.** Build the November calendar per SO under the rules in section 3: 6 dealer visits a day with time held for 2 demand-generation calls. High-value de-growing dealers go early in the month.
- **Route.** Sequence each day to minimise travel and fit the working day. Any method is fine (OR-Tools, OSRM, Google Maps, or straight-line distance with stated speed assumptions), but it must handle a dense city and a spread-out upcountry territory.

- **Nudge.** For at least five scheduled visits, write the WhatsApp message the SO would get the evening before (under 300 characters), grounded in the data: for example points needed for the next slab, a recent drop, or the product tier the dealer has stopped buying.
- **Re-plan.** Write a command that, each evening, takes what happened that day (missed visits, weather, each dealer's sales) and rewrites the rest of the month. A sharp sales drop should trigger a visit; a sales hike can cancel one; rain or traffic tomorrow means only 5 dealer visits will happen, so put the important ones first. Never drop club coverage, and log your reasons.

## 3 Constraints

| Rule | Detail |
|---|---|
| Planning month | 1–30 November 2026. Monday to Saturday are working days (25 days); Sundays are off. See `Planning_Calendar`. |
| Focus dealers | Choose each SO's focus dealers from the `Dealers` sheet. Capacity is 6 dealer visits × 25 days = 150 visits a month, so the focus set must fit once club dealers take two slots each. |
| Visit frequency | Loyalty-club dealers: **2 visits a month**, about two weeks apart. Other focus dealers: **1 visit a month**. |
| Calls per day | Every SO, every working day: **6 focus-dealer visits + 2 demand-generation calls** (architects, designers, contractors or carpenters). No demand-generation data is provided: do not plan those accounts, but **reserve enough time in each day for the 2 calls**, including travel. |
| Working day | Assume 09:30–18:30 from the SO's day-start point (`Officers`). A productive call lasts at least 10 minutes; observed medians are 22–36 minutes. State your travel-speed assumptions. |
| Guardrails | **No dealer gets more than 2 visits a month.** Club dealers get exactly 2, about two weeks apart; other focus dealers get 1. Separately, flag history that breaks the rules: calls under 10 minutes and dealers visited far more often than needed. |
| Partial data | The data is incomplete by design: some dealers have no location or no usable GST record, demand-generation accounts are absent, and parent/child outlets share an owner. We expect you to make progress with what is there, handle gaps explicitly and state your assumptions, not to wait for perfect data. |

## 4 Data provided (one Excel workbook)

| Sheet | Contents |
|---|---|
| `Officers` | SO-A and SO-B (dense Bengaluru territories), SO-C (Hubballi–Dharwad–Belagavi, upcountry); approximate day-start points. |
| `Dealers` | 677 outlets across the three SOs, with location (395 located), loyalty-club status and tier, parent outlet, town and source flags. |
| `Sales_Daily` | Secondary sales per dealer per day per product tier, April 2025 to 15 October 2026 (about 40,500 rows). Returns are negative. |
| `Product_Tiers` | Scheme points per sheet for each product tier (8 / 4 / 2 / 1 / 0). |
| `Scheme_Targets`, `Scheme_Rules_NonClub` | Slab targets in points for club dealers; monthly and annual slabs for everyone else. |
| `SFDC_Activities` | Every visit record for the three SOs, April–October 2026: planned (PJP) or ad hoc, status, check-in/out times, account and location (about 3,700 rows). Past demand-generation calls are included to show how a real day is spent. |
| `GST_Filings` | Public GST profile per dealer: turnover band (exact slab where available), e-invoicing status, filing frequency and punctuality, compliance colour, nature of business. Some dealers have no usable record. |

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| `Baseline_Metrics` | How the current manual plan performs, per SO per month: your benchmark. |

IDs are pseudonymous; names, GSTINs, phone numbers and notes are removed. Coordinates carry small random offsets. Sheet and rupee figures are scaled by a fixed factor, so compare them rather than reading them as exact.

## 5 Success metrics: we simulate November against your plan

We play November out day by day with the simulator in `PJP_Simulator.zip`. Each day your SOs execute the plan in your sequence with real travel time and the 2 demand-generation calls held back; **rain or traffic caps a day at 5 dealer visits**; some dealers are unavailable; each dealer's sales are simulated from its history, with hidden **sales drops** (a visit within 7 days recovers them) and **sales hikes** (a visit can be cancelled). Every evening the simulator calls your re-plan command. You can run it yourself on practice seeds 1–3; we score on unpublished seeds.

| Metric | Definition | Weight |
|---|---|---|
| **Simulated adherence** | Visits completed ÷ visits in the plan on the morning of each day. Today's manual plans reach about 60%; the target is 90%. | 30% |
| **Focus-dealer coverage** (primary business metric) | Focus dealers in your original plan who received their required visits in the simulation (club 2, others 1). | 25% |
| **Drop response** | Sales-drop events visited within 7 days of onset. | 20% |
| **Club cadence** | Club dealers visited twice, at least 10 days apart. | 10% |
| **Hike handling** | Visits cancelled rather than made while a dealer's sales were spiking. | 5% |
| **Guardrails and reliability** | No day above 6 dealer visits, no dealer above 2 visits, no visits to another SO's dealers, no re-plan failures or timeouts (120 s). | 10% |

Scores are averaged over the hidden seeds. The simulated score informs, but does not replace, the review in section 7: a clear, well-reasoned approach with a slightly lower score beats an opaque one tuned to the simulator.

## 6 Deliverables

- **Code** in a Git repository (Python preferred), runnable end to end from the workbook with one command, with a README.
- **The November plan** as a CSV per SO: `date, so_id, sequence, account_id, account_type, planned_arrival, travel_min_from_previous, reason`.
- **A re-plan command** that meets the simulator contract in the simulator README (`--state --plan --out`, under 120 seconds), plus your `summary.json` for practice seeds 1–3.
- **Five or more WhatsApp nudges**, each with the data behind it.
- **A two-page write-up:** approach, assumptions, the section 5 metrics against today's baseline, what failed or was hard, and what you would do with more data or time. A map of one day per SO is welcome.

## 7 How we evaluate

| Area | Weight | What good looks like |
|---|---|---|

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|---|---|---|
| Problem framing and judgement | 20% | Sensible assumptions, honest about data gaps, metrics tied to the business problem. |
| Ranking logic | 25% | Value-weighted, explainable, uses GST potential, handles dormant and parent/child outlets. |
| Scheduling, routing and simulated results | 25% | Rules respected, feasible days, works in both metro and upcountry. |
| Code quality and reproducibility | 15% | Clean, tested where it matters, one-command run. |
| Communication | 15% | Clear write-up and nudges a Director of Sales would act on. |

> **Ground rules.** Submit within five days of receiving this. Use any tools, AI included, but be ready to explain every line. The data is confidential: use it only for this assignment and delete it afterwards. Questions are welcome by reply email; state any assumption you make rather than waiting.